Annotated slides. Laptop failed in class, so notes here might be a little different from those given in class.

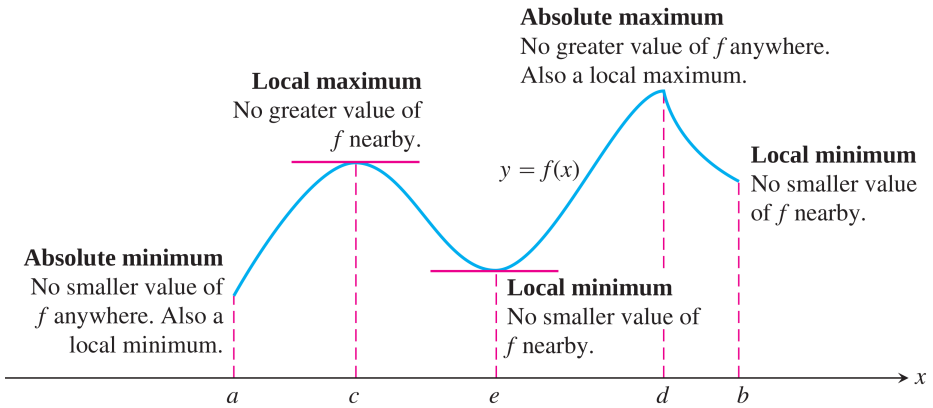
2627-MA140 Engineering Calculus

Week 06, Lecture 1
Logarithmic Differentiation; Max & Min

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University of Galway

Tuesday, 13 October, 2026



Info: Survey, Assignments, etc

- The module survey for MA140 has started. Please take a few minutes to complete it. See https://universityofgalway.instructure.com/courses/58477/discussion_topics/252559
  - It only takes a few minutes
  - We take in the input seriously, and will update you on the main findings and the actions we will take.
  - Try to mix positive comments (so we know what we are doing well) with suggestions for improvements.
- **Problem Set 4** is open and is due next Monday, 19th October at 18:00.
- **Problem Set 5** will follow at the end of the week.
- Grades for Class Test 1 were posted Wednesday,

About tomorrow's lecture...

Industrial action, Wednesday, 14th October

- Due to Public Services Unions being on strike tomorrow, unless it is cancelled, there will be no on-campus teaching activities tomorrow.
- We will adjust to module content accordingly.
- There will be no disadvantage to you. Specifically all topics on the exam paper will have been covered in class as usual.

Today, we'll max out on...

1. Recall: Logarithms
2. Logarithmic differentiation
3. Higher-order Derivatives
4. Maxima and minima
   - Overview
   - Critical points
5. The First Derivative Test
   - Increasing/decreasing
   - Derivatives
   - Example
   - The test
6. Exercises [Revise later!]

Slides are on canvas, and at niallmadden.ie/2627-MA140



Text-book references: End of 3.9 for Logarithmic Differentiation. Section 4.3 for Maxima and Minima. Link to book: https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)

Recall: Logarithms

Last week we learned that:

- The number $e \approx 2.7182818$ is special for lots of reasons, including that, if we define the exponential function $f(x) = e^x$, then $f'(x) = f(x)$.
- $f(x) = e^x$ is sometimes called the *natural exponential function*.
- **Logarithmic functions** can be thought of as the inverse of the corresponding **exponential** function:
  If $y = b^x$ then $\log_b(y) = x$.
- The **natural logarithmic function** can be written as $\log_e(x)$ or $\ln(x)$.
- $\frac{d}{dx} \ln(x) = \frac{1}{x}$

Euler's Number

ln (x)

Log ↗ ↖ natural .

Logarithmic differentiation

We can use logarithms when computing derivatives: **Logarithmic differentiation**, which helps us differentiate functions with x, or a function of x in the exponent, such as $y = (2x)^{\sin(x)}$ or $y = x^x$.

Strategy:

- Take $\ln()$ of both sides
- Simplify, using properties of logarithms.
- Differentiate.
- Solve for $\frac{dy}{dx}$

$\log(ab) = \log(a) + \log(b)$

$\log(a^b) = b\log(a)$.

usually using Chain Rule.

Logarithmic differentiation

Example

Differentiate $f(x) = x^x$. (with respect to x)

Set $y = x^x$. Take the natural log of this.

$\ln(y) = \ln(x^x) \Rightarrow \ln(y) = x\ln(x)$ since $\ln(a^b) = b\ln(a)$

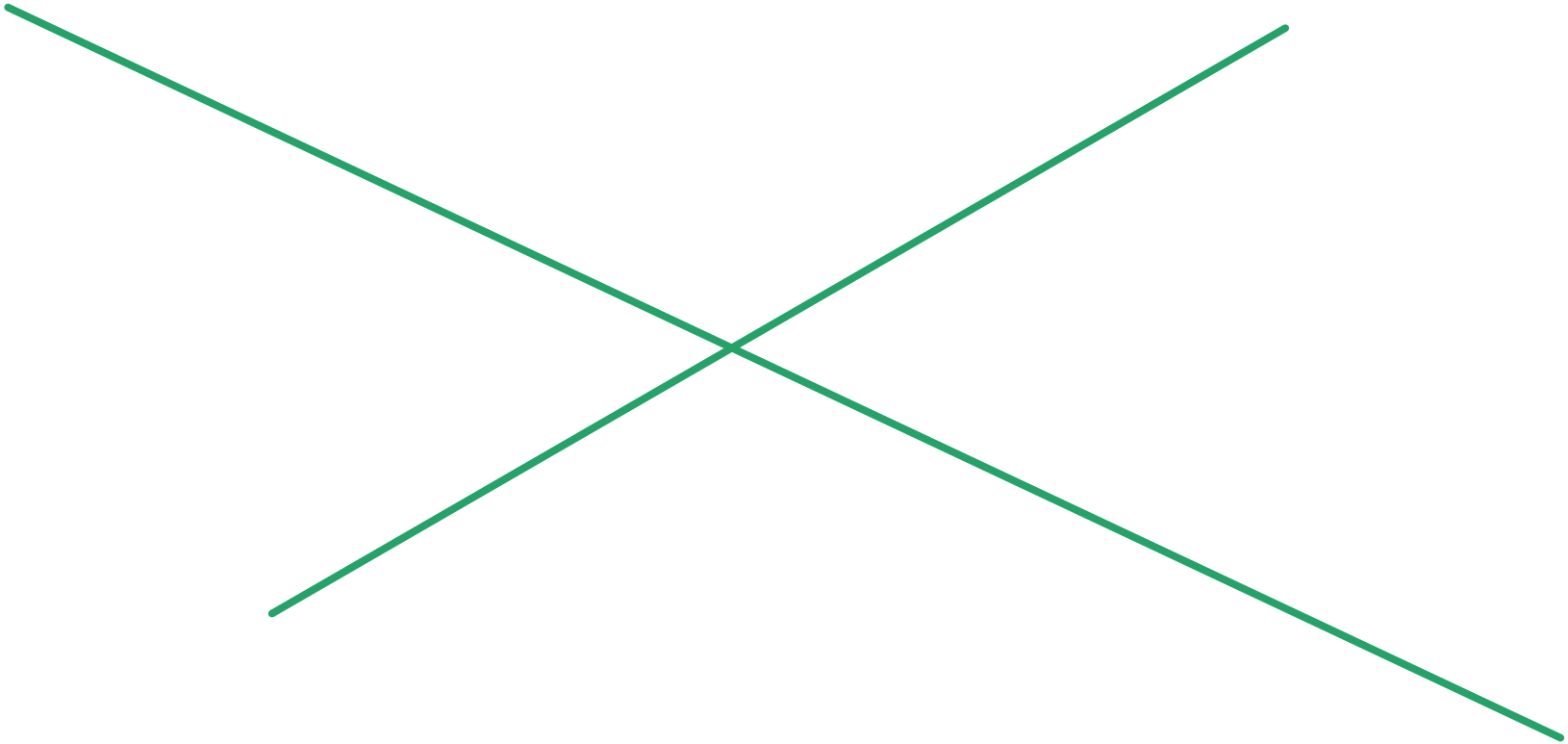
Left side: $\frac{d}{dx}(\ln(y)) = \frac{d}{dy}(\ln y)\cdot\frac{dy}{dx} = \frac{1}{y}\frac{dy}{dx}$.

Right side: $\frac{d}{dx}[x\ln(x)] = x\cdot\frac{1}{x} + \ln(x)(1)$ (by Product rule)

So $\frac{1}{y}\frac{dy}{dx} = 1 + \ln(x)$. This gives $\frac{dy}{dx} = y(1+\ln(x))$.

To finish, use $y = x^x$, and we get $\boxed{\frac{dy}{dx} = x^x(1+\ln(x))}$

Logarithmic differentiation



Higher-order Derivatives

We know that the derivative of $f(x)$, denoted $f'(x)$, is itself a function.

That implies that $f'(x)$ can itself be differentiated, which is called the **second derivative** of f. It is denoted as

$$\frac{d^2f}{dx^2} \quad \text{or} \quad f''(x) \quad \text{or} \quad f^{(2)}(x).$$

We can continue this process to get higher-order derivatives as long as the preceding derivative is again differentiable.

The first and second derivatives f' and f'' (if they exist) provide valuable information about the function and its graph, particularly concerning local or global maxima, local/global minima and points of inflection.

Higher-order Derivatives

Example

Find the **second** derivative of the functions

(i) $f_1(x) = 1 - 4x$ (ii) $f_2(x) = x^2$ (iii) $f_3(x) = e^x$

(i) $f_1(x) = 1 - 4x$ so $f'(x) = 0 - 4 = -4$. Then $f''(x) = \frac{d}{dx}(-4) = 0$.

(ii) $f_2(x) = x^2$. So $f'(x) = \frac{d}{dx}(x^2) = 2x$. Then $f''(x) = \frac{d}{dx}(2x) = 2$

(iii) $f_3(x) = e^x$. So $f'(x) = e^x$ and $f''(x) = e^x$.

Higher-order Derivatives

Example

Find the **second** derivative of the functions

(i) $f_4(x) = \ln(x)$ (ii) $f_5(x) = \sin(x)$

(i) $f_4(x) = \ln(x)$. $f_4'(x) = \frac{d}{dx}(\ln(x)) = \frac{1}{x} = x^{-1}$.

Then $f''(x) = \frac{d}{dx}(x^{-1}) = (-1)x^{-2} = \frac{-1}{x^2}$

(ii) $f_5(x) = \sin(x)$ So $f'(x) = \cos(x)$, so $f''(x) = -\sin(x)$.

This section of MA140 is concerned with using techniques of differentiation to finding where a function is

- Increasing
- Decreasing
- Has its maximum value
- Has its minimum value

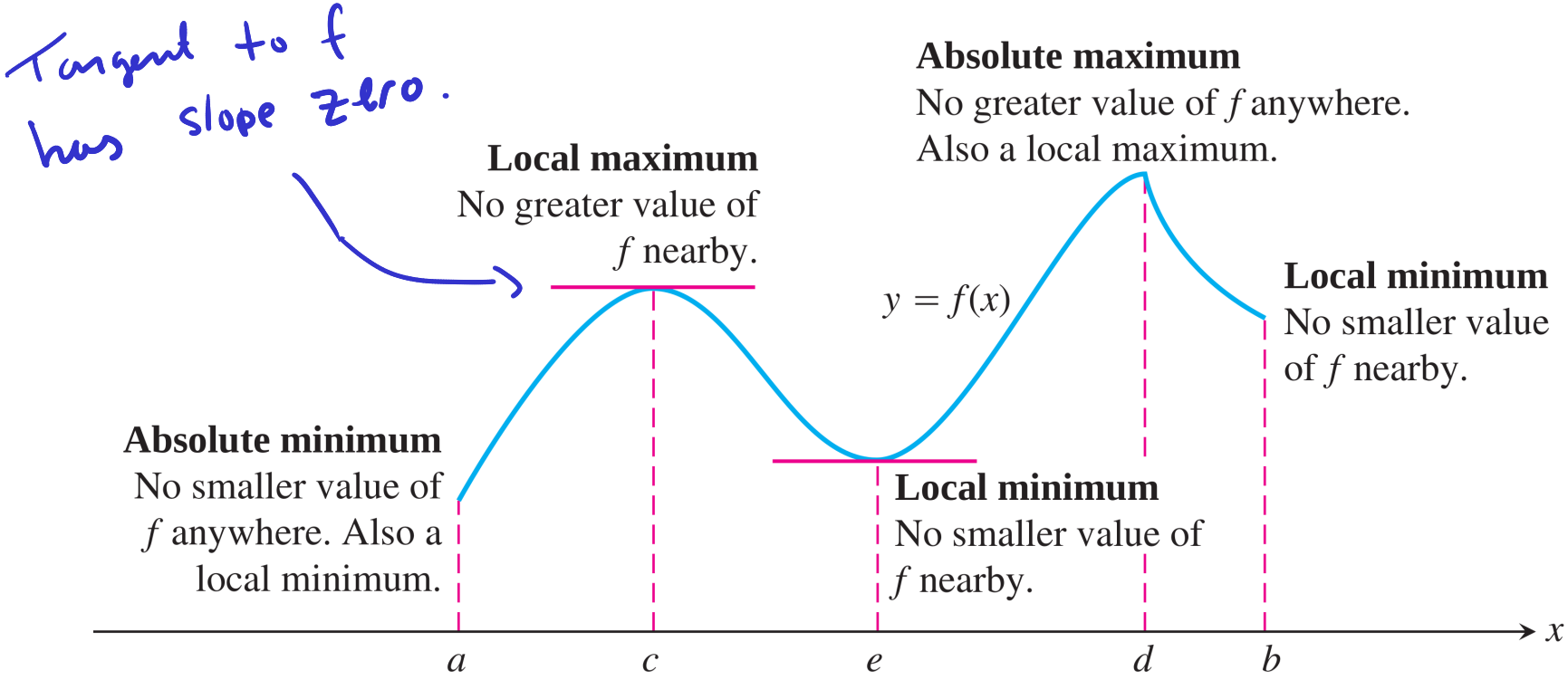
Along the way we'll learn about **critical values** and the **first derivative test**.

Mathematical English

- The plural of **maximum** is **maxima**;
- The plural of **minimum** is **minima**;
- An **extremum** a maximum or a minimum.
- The plural of **extremum** is **extrema**.

Maxima and minima

Given an interval $[a, b] = \{x \in \mathbb{R} \mid a \leqslant x \leqslant b\}$, consider the function $f : [a, b] \to \mathbb{R}$ whose graph is given below. It illustrates local and absolute (="global") maxima and minima. Collectively, these are called **extrema**.



Definition: critical points

Let c in an point in the domain of a function f. We say that $x = c$ is a **critical point** of $f(x)$ if either

$$f'(c) = 0 \qquad \text{or} \qquad f'(c) \text{ does not exist.}$$

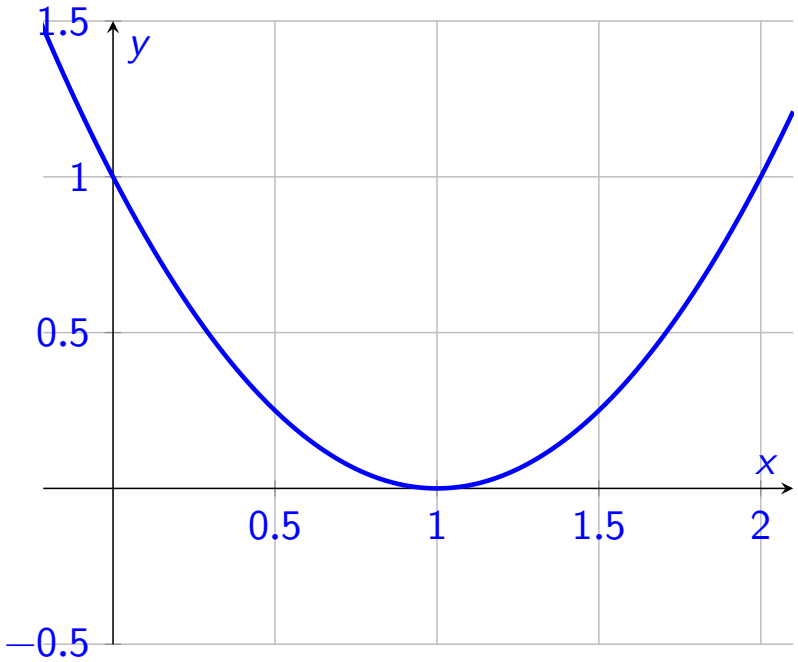
Important: If f has an extremum at $x = c$, then c must be a critical point of f (This is called "Fermat's Theorem").

So, to find a maximum or minimum of f, it is enough to check at the critical points.

Warning: All extrema are at critical points, but not all critical points correspond to a extrema.

Example

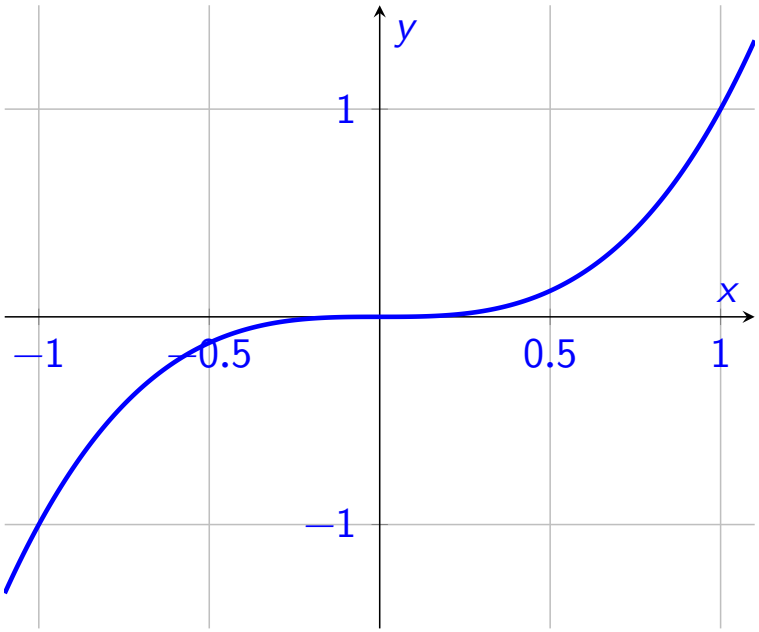
$f(x) = x^2 - 2x + 1$ has one critical point. Find it. Does it correspond to an extremum?



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Example

Find all critical points of $f(x) = x^3$. Do they correspond to extrema?



Definition (Increasing/Decreasing)

Let f be a function whose domain includes the interval $[a, b]$. Let let x_1 and x_2 be any two points in $[a, b]$ with $x_1 < x_2$.

- If $f(x_1) < f(x_2)$, then f is said to be *increasing* on $[a, b]$.
- If $f(x_1) > f(x_2)$, then f is said to be *decreasing* on $[a, b]$.

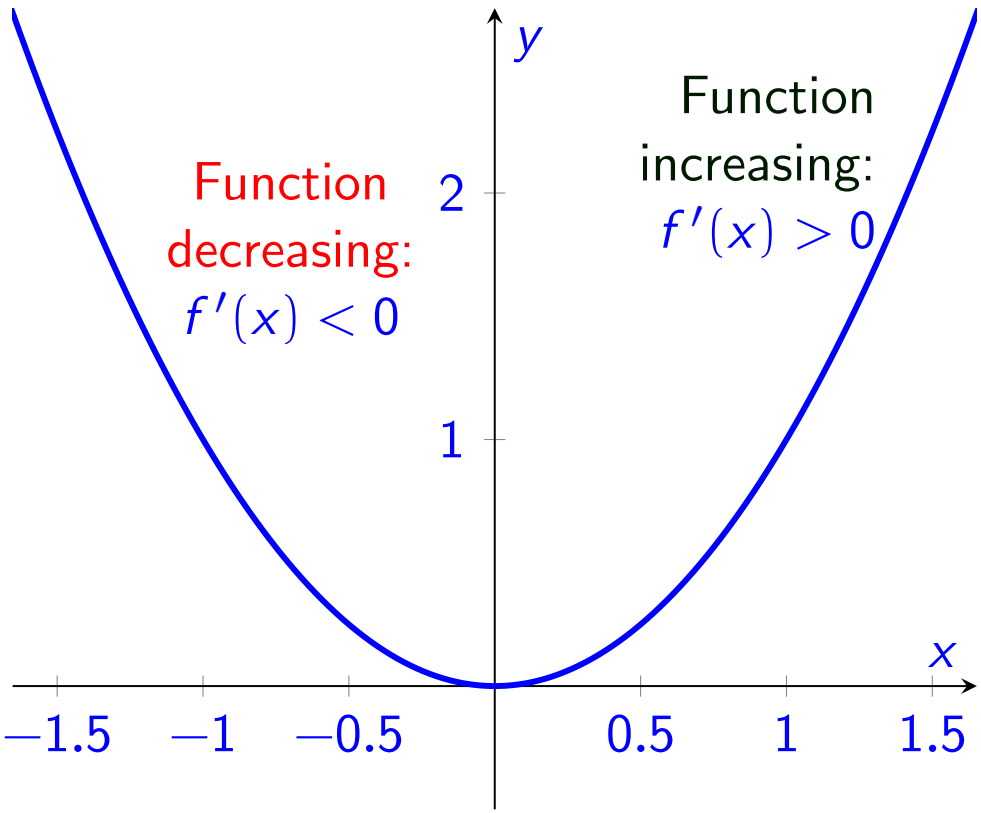
Example

The function $f(x) = x^2$ is decreasing on $(-\infty, 0]$, and increasing on $[0, \infty)$.

Theorem

Suppose that f is differentiable on an interval $[a, b]$.

- If $f'(x) > 0$ at each point $x \in [a, b]$, then f is increasing.
- If $f'(x) < 0$ at each point $x \in [a, b]$, then f is decreasing.



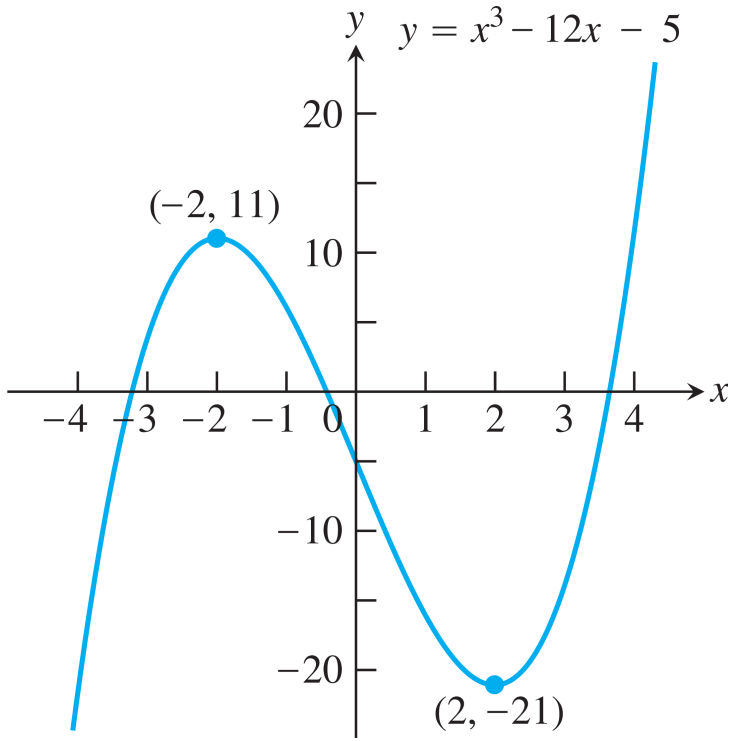
Example

Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the intervals on which f is increasing and decreasing

Idea: find $f'(x)$ and then solve for $f'(x) = 0$.

The critical points $c = -2$ and $c = 2$ of $f(x) = x^3 - 12x - 5$ subdivide the domain of f into intervals $(-\infty, -2), (-2, 2)$ and $(2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f at a convenient point in each subinterval.

	-2		2	
$3(x+2)$	$-$	$\bullet$	$+$	$+$
$x-2$	$-$		$-$	$\bullet$ $+$
$f'(x)$	$+$	$\bullet$	$-$	$\bullet$ $+$



Important:

- If $f(x)$ has a local minimum of $f(x)$ at $x = c$, then it switches from **decreasing** to **increasing**. That means, $f'(x)$ changes sign at $x = 2$. Therefore, $f'(c) = 0$.
- If $f(x)$ has a local maximum at $x = c$, we have that $f'(c) = 0$.

First Derivative Test for local maxima and minima

Suppose that c is a critical point of a differentiable function f.

- If f' changes from negative to positive through c, then f has a local minimum at c.
- If f' changes from positive to negative through c, then f has a local maximum at c.
- If f' does not change sign through c (that is, f' is positive on both sides of c or negative on both sides), then f does not have a local maximum or minimum at c.

Exercises

Exercise 6.1.1

Let $f(x) = x^2 e^x$. Find $f'(x)$, $f''(x)$ and $f'''(x)$.

Exercise 6.1.2 : From 23/24 Exam

Let $f(x) = \ln(x^2 + 1)$.

(i) Find all critical point(s) of f and determine whether f has a local minimum, local maximum or neither.

(ii) Determine the interval on which f is increasing.

(iii) Determine the interval on which f is decreasing.